

Mohamed Abdulhamid Tahawy

AI Engineer | Applied Machine Learning, LLMs & Multimodal Systems

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Profile Summary

Results-driven **AI Engineer & Technical Instructor** with a B.Sc. in Computer Science & Artificial Intelligence (Grade: **Very Good**) and certified as an **AWS Machine Learning Engineer - Associate (MLA-C01) & AWS AI Practitioner**. Hands-on expertise building end-to-end applied AI solutions, multimodal pipelines, and intelligent conversational systems. Proven track record architecting enterprise RAG platforms with ChromaDB, OCR-powered document intelligence engines (EduBot), and multi-turn conversational agents with stateful memory, alongside algorithmic trading automation systems achieving an **89.5% win rate**. Proficient in full-stack AI prototyping using Python, PyTorch/TensorFlow, REST APIs, and Docker/Nginx microservice deployments on Linux VPS. Experienced technical leader who designed and delivered a 90-hour Machine Learning curriculum at NTI, mentoring engineers on model evaluation, reproducible MLOps workflows, and production best practices. Native Arabic speaker equipped with strong technical and cross-functional communication skills.

Technical Skills

- **Multimodal AI & Document Intelligence:** OCR Pipelines (Tesseract, OpenCV), Scanned Document Parsing, Forms & Table Extraction, Computer Vision (YOLO, OpenCV, Face Recognition, Image Captioning), Speech/Audio Concepts (STT / TTS Pipelines), Multimodal Fusion.
- **Large Language Models & NLP:** Retrieval-Augmented Generation (RAG), Foundation Models (Gemini API, OpenAI), Arabic NLP & Dialect Processing, Vector Databases (ChromaDB), Semantic Search, Question Answering, Text Summarization, Prompt Engineering, Agent Memory.
- **Machine Learning & Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, CNNs, RNNs, Transformers, Model Optimization & Quantization, Model Evaluation (Accuracy, Precision, Recall, F1, RMSE, BLEU, Latency).
- **Backend Engineering & Deployment:** Python, REST APIs, FastAPI, Flask, Docker Containerization, Nginx Reverse Proxy, PM2 Process Management, Linux / VPS Server Administration, Streamlit Dashboards, Git/GitHub, CI/CD Principles.
- **Cloud & System Architecture:** AWS Cloud Infrastructure (Certified Machine Learning Engineer - Associate MLA-C01, AI Practitioner, Cloud Practitioner), ChromaDB Vector Store, SQL, Neo4j Graph Database, Microservices Architecture.

Professional Experience

AI Engineer Intern | Abs.ai

Nov 2025 – Dec 2025 | New Cairo, Egypt

- **Document Intelligence & OCR (EduBot):** Engineered an end-to-end document understanding solution that extracts, parses, and structures text and tabular content from physical/scanned textbooks via OCR; indexed content into a searchable vector database deployed on a VPS.
- **Arabic NLP & Enterprise RAG Assistant:** Co-developed a bilingual banking assistant scraping and preprocessing retail, corporate, and SME data from NBE and Banque Misr; utilized ChromaDB for vector retrieval and Gemini API with user-resettable JSON memory.
- **Algorithmic Trading Bots:** Built and optimized algorithmic forex trading bots on MT5, improving strategy win rate from 38% to 89.5% through iterative backtesting, feature engineering, and live testing on VPS servers with static IPs.
- **Production Deployment & DevOps:** Containerized AI pipelines with Docker and deployed web services with Nginx reverse proxy and PM2 on Linux VPS, configuring domain routing, SSL certificates, and server telemetry.

AI Instructor & Technical Lead | National Telecommunication Institute (NTI)

Nov 2025 – Present | Giza, Egypt

- **Curriculum Architecture & Delivery:** Designed and led a 90-hour intensive program on End-to-End Machine Learning and Applied AI for Business for 50+ undergraduate engineering and computer science students.
- **Practical Mentorship & MLOps:** Guided students through hands-on deployment workflows using cloud computing (GCP/Kaggle), REST API integration, and industry-standard model evaluation metrics (Precision, Recall, F1, RMSE).
- **Applied Case Studies:** Mentored real-world business case studies, establishing standard engineering practices for clean code, model versioning, API documentation, and stakeholder reporting.

AI & Cybersecurity Engineering Intern | NTI

Jul 2025 – Oct 2025 | Smart Village, Egypt

- **Threat Prediction & Anomaly Detection:** Developed machine learning models to detect anomalies and predict cyber threat vectors across high-volume network telemetry logs.
- **Automated Mitigation Workflows:** Engineered automated defense pipelines triggering dynamic firewall rules to block malicious IPs upon classification, achieving sub-second detection latency (~200 ms).

Machine Learning Intern | Codsoft

May 2024 | Remote

- **Applied AI Prototyping:** Trained and validated supervised ML models for predictive classification and regression; packaged inference models into lightweight REST API endpoints with structured documentation.

Selected Applied AI & Multimodal Projects

FlowWatch-AI: Real-Time Network Threat Intelligence Platform | NTI Capstone

Sep 2025

- **Real-Time Architecture & Defense:** Engineered a real-time monitoring system integrating multi-stage ML classification with live traffic telemetry; achieved sub-second detection (~200 ms) with automated IP defense and structured JSON audit logging.
- **Interactive Dashboard & Technologies:** Built an interactive Streamlit dashboard and Power BI visualizations for live traffic analysis and threat auditing. Tools: Python, Scikit-learn, Streamlit, Power BI, REST APIs, JSON Logging.

I-Monitor: Multimodal Image Understanding & Captioning System | DEPI Capstone

Oct 2024

- **Vision-to-Language Deep Learning:** Developed an assistive AI solution utilizing deep learning image-captioning models trained on COCO 2017 to translate visual scenes into natural language descriptions for visually impaired users. Tools: TensorFlow, Computer Vision, CNN-LSTM/Transformer, Python.

Workplace Safety & Employee Monitoring Solution | Client Freelance Project

Feb – Apr 2025

- **Computer Vision & Behavior Analysis:** Designed and deployed an AI-powered tool assisting company managers in monitoring employee behavior and ensuring workplace safety. Tools: Python, YOLO, OpenCV, Face Recognition.

WSAM Smart Management & Biometric Platform | University Capstone

Jun 2024 | Grade: Excellent

- **Biometric Security & Automation:** Engineered a smart facility management prototype featuring CCTV stream analytics, automated alerting, and multi-factor biometric authentication. Tools: Python, OpenCV, Face Recognition, Multi-threading.

Education

B.Sc. in Computer and Artificial Intelligence | Fayoum University

2020 – 2024 | Fayoum, Egypt

- **Academic Standing:** Major in Computer Science; Overall Grade: **Very Good**. Graduation Capstone Grade: **Excellent**.
- **Relevant Coursework:** Natural Language Processing, Computer Vision, Machine Learning & Deep Learning, Distributed Systems, Software Engineering, RESTful APIs, Database Design, Algorithms & Data Structures.

Certifications & Professional Training

- **Industry Certifications:** AWS Certified Machine Learning Engineer - Associate (MLA-C01) • AWS Certified AI Practitioner (Oct 2025) • AWS Certified Cloud Practitioner (Feb 2025) • Huawei HCCDA-AI (Oct 2025) • NVIDIA DLI (Sep 2025).
- **Specialized Professional Training:** AI in Cybersecurity (520 hrs, NTI) • AWS Machine Learning Trainee (182 hrs, DEPI.gov) • Machine Learning Specialization (80 hrs, Coursera by Andrew Ng) • Quality Control & Assurance (40 hrs, USAID/UCCD).

Additional Information

- **Languages:** Arabic (Native Speaker), English (B2 / Professional Working Proficiency).
- **Extracurricular Leadership:** Machine Learning Member & 'Best Member of the Month' (Nov 2023), Google Developer Student Clubs (GDSC-FU).
- **Status & Availability:** Military Exempted; Available for employment. Location: Giza / Cairo, Egypt.